

Surgical Management of Cancer Patients

(Colon& Urinary)

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Transitional cell tumours

Key facts

- Transitional cell tumours (TCT) may affect any part of the urinary epithelium (renal pelvis, ureter, bladder, or very rarely, urethra).
- TCT have a spectrum of disease from **benign** superficial 'papilliferous' growths to frankly **invasive** transitional cell carcinoma (TCC);
- progression may occur from benign to more malignant forms with time.
- TCC of the bladder. **Fifth** commonest cause of cancer deaths and the commonest form of bladder cancer in the UK; ♂:♀, 3:1.

TCT

TCT is associated with:

- **Exposure to aromatic hydrocarbons**, e.g. workers in the petrochemical, industrial dye, rubber industries, chimney sweeps.
- **Smoking** (especially in women).
- Risk is probably due to excretion of carcinogenic products excreted and concentrated in the urine and tumours are more likely in locations exposed to urine for the longest periods, i.e. bladder.

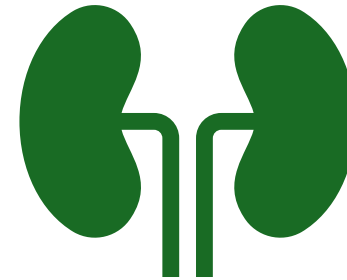
Pathological Features

- The majority (70%) are **superficial** in nature at diagnosis, being confined to the mucosa.
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- **Invasion** into the lamina propria, muscle, and perivesical fat can occur with lymphatic and distant spread occurring in advanced cases
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- **TCC *in situ*** is pre-invasive and associated with a high risk of muscle invasive disease if not adequately treated
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- **Squamous cell carcinoma.** Usually caused by chronic irritation due to **schistosomiasis** infestation (bilharzia), indwelling catheter, repeated previous surgical interventions.
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- **Adenocarcinoma.** Rare; presents in middle age and is usually located in the dome of the bladder in association with the urachus

Clinical Features



- The majority of cases present with **painless haematuria**.



- Other features are **painful micturition, renal colic** due to blood clot, disturbance of urinary stream, retention of urine.

Diagnosis and Investigations

1. Urine cytology; May reveal malignant cells; if there are malignant cells, TCC or carcinoma *in situ* will probably be present.

2. Cystoscopy:

- Usually carried out using a fiberoptic flexible cystoscope and local anaesthetic gel.

3. Upper tract imaging

- Used to identify and assess pelviureteric tumours.

A. IVU or Ultrasound scan.

- Ultrasound scan permits examination of the renal cortex and will detect tumours of 1cm diameter in the pelvicalyceal system, ureter, and bladder.

B. MRI and CT scanning to detect local or systemic spread.



Treatment

1. Superficial TCT

- Remove; completed by **endoscopic resection**.
 - **Intra-vesical chemotherapy** reduces the risk of tumour recurrence (single dose of mitomycin C instilled after resection of the tumour).
 - For multiple or recurrent TCC, six intravesical treatments are given.
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Treatment

2. Carcinoma in situ

Requires thorough therapy to prevent invasive TCC.

- Immunotherapy with intravesical BCG is effective in 60% of cases.

3. Invasive TCC

- Curative therapy can be offered with **radical cystectomy** (*combined with a urinary diversion via an ileal conduit*) or radical radiotherapy.

4. Squamous cell and adenocarcinoma

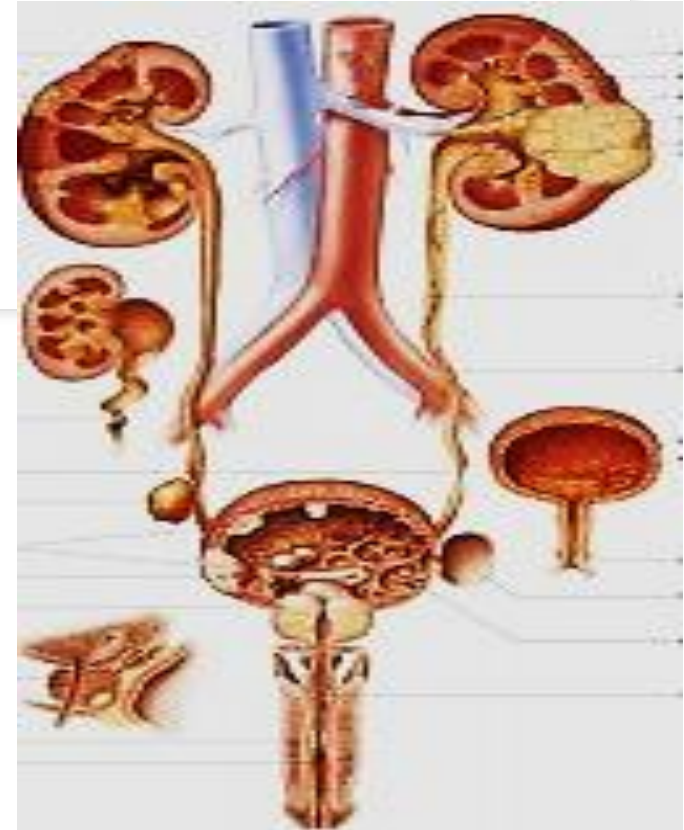
- **Radical cystectomy** provided general condition allows.
- Usually **resistant** to radiotherapy and chemotherapy.

Transurethral Resection

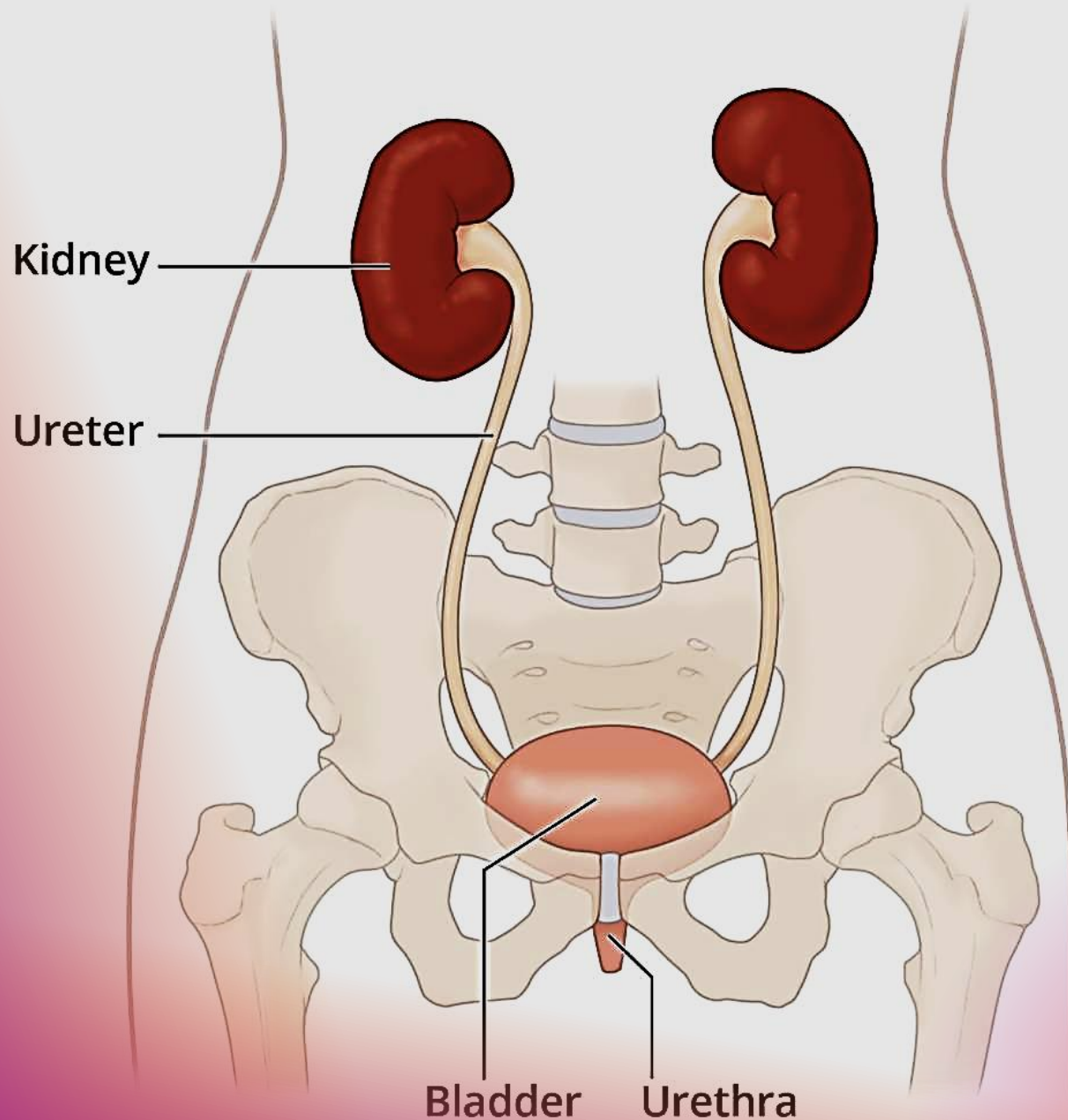
- Usually carried out using **a rigid endoresectoscope** under GA.
 - Permits resection of all or part of the tumour using a **diathermy 'loop'**, with the tumour resected piecemeal; resection may be carried out into deep tissue (the muscle wall of the bladder beneath the tumour).
 - Subsequent pathological examination will determine the histological grade and the pathological stage, e.g. depth of invasion.
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What is urinary diversion?

- Urinary diversion is a surgical procedure that creates a new way for urine to exit the body when urine flow is blocked or when there is a need to bypass a diseased area in the urinary tract.
- The urinary tract is the body's drainage system for removing urine, which is made of wastes and extra fluid.
- Urinary tract is designed to have the urine flow from the kidneys, through the ureters, to the bladder, and out the urethra.
- When the urine can't flow normally, it may build up in the bladder, ureters, or kidneys.
- This buildup of urine may cause pain, urinary tract infections, urinary stones or calculi, damage to the urinary tract, or kidney failure.



Urinary Tract



What are the types of urinary diversions?

- **The main types of urinary diversion include:**

1. Bladder catheterization
2. Cystostomy
3. Nephrostomy
4. Ureteral stent
5. Urostomy
6. Continent urinary diversion

Bladder Catheterization

- Bladder catheterization involves inserting a thin, flexible tube—called a catheter—into the bladder to drain urine.
- The urine drains into a collection bag outside the body.

Types of urinary catheters:

- **Foley catheter**, inserted into the bladder through the urethra
- **suprapubic catheter**, inserted into the bladder through a small hole in the skin beneath the umbilicus.

Duration:

- Urinary catheters may remain in place for several days or weeks while tissues heal after urinary tract surgery or treatment of urinary blockage and, in some cases, may be permanent.
- Catheters that are in place for longer periods of time need to be replaced with a new catheter periodically.

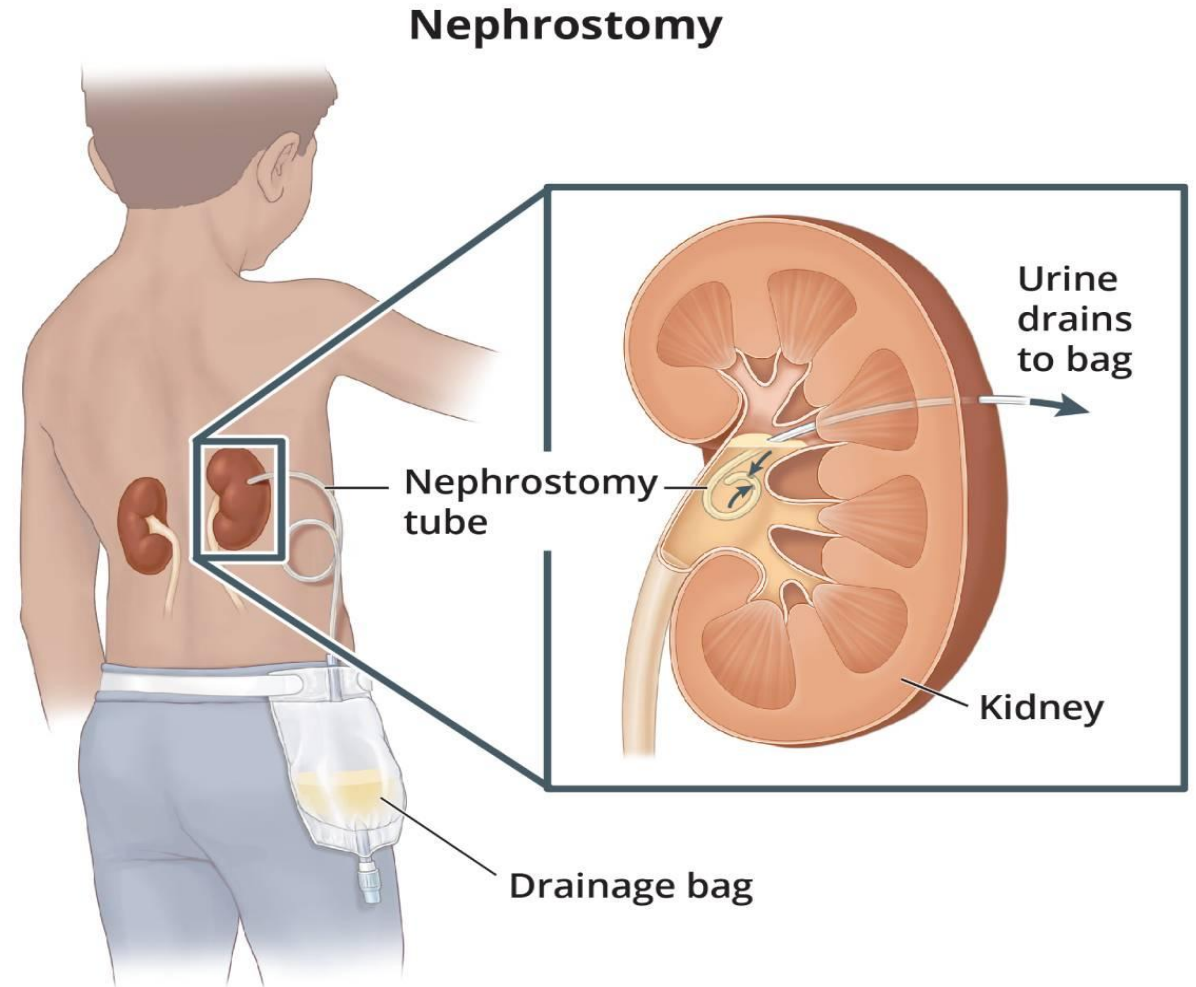
Cystostomy & Nephrostomy

Cystostomy

A cystostomy is a surgical procedure where a small tube is inserted into the bladder through the skin of the lower abdomen. The tube allows urine to drain from the bladder into a bag outside the body.

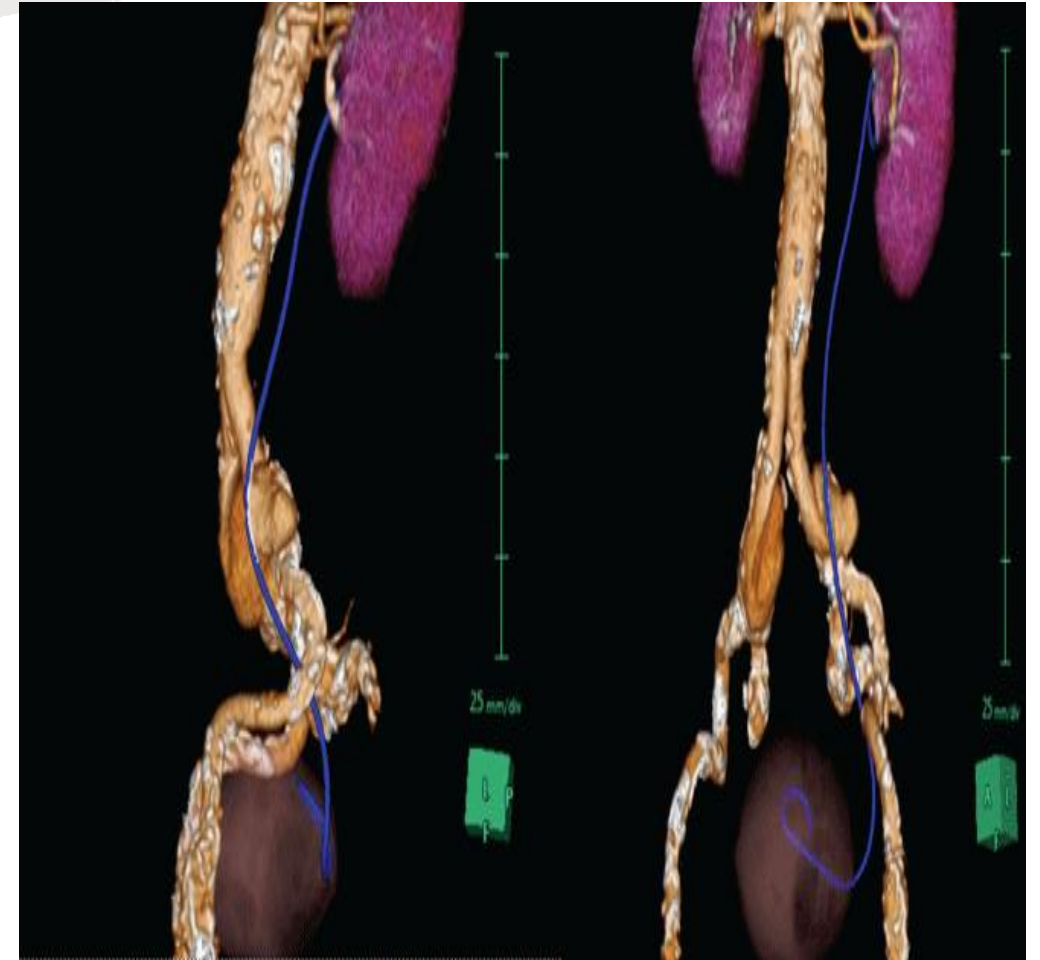
Nephrostomy

- A surgeon or radiologist makes a tiny incision and inserts a small tube, called a nephrostomy tube, through the skin of the back into the kidney.
- The nephrostomy tube allows urine to drain from the kidney into a bag outside the body.
- You may need a nephrostomy when being treated for a kidney stone or when your ureters are narrowed, blocked.
- Depending on the reason for the nephrostomy and how quickly the body heals, the nephrostomy tube may be used for different lengths of time.

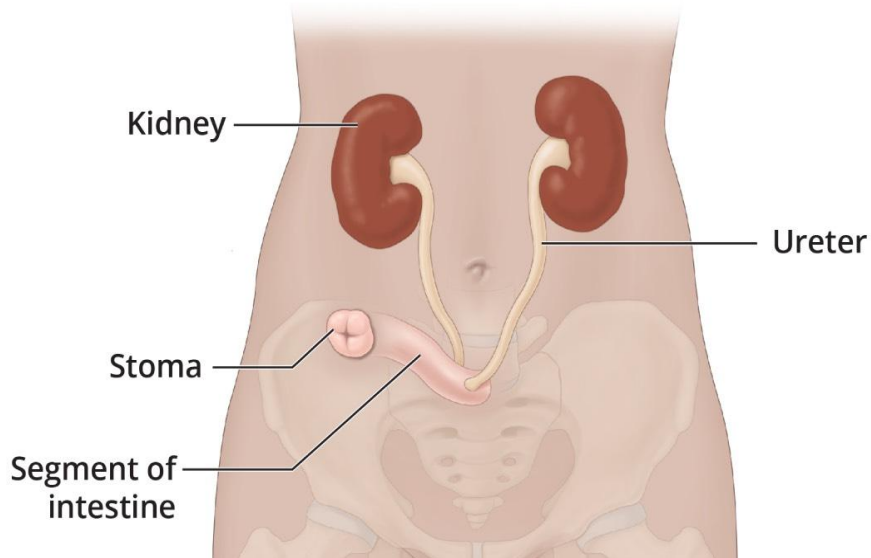


Ureteral Stent

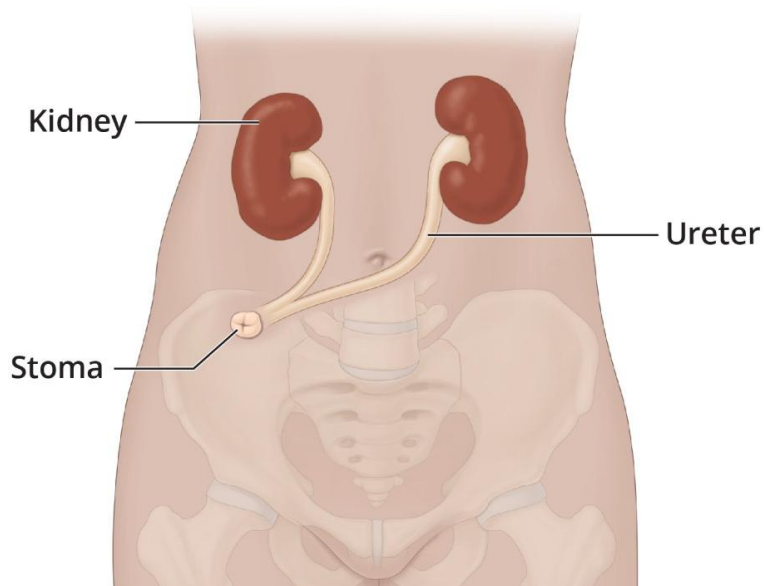
- A ureteral stent is a thin flexible tube that is inserted into the ureter to help urine flow from the kidney to the bladder.
- The ureteral stent is guided with a [cystoscope](#) into the ureter, then one end of the stent is placed in the kidney and the other end is placed in the bladder.
- You may need a ureteral stent if one of the ureters is blocked as a result of surgery, a kidney stone, a tumor, or infection.
- A ureteral stent is usually **temporary** but, in some cases, can be used to permanently manage a blockage of the ureter.
- Ureteral stents that are in place for longer periods of time need to be replaced periodically.



Ileal Conduit with Stoma



Cutaneous Ureterostomy



Urostomy

- A urostomy is a [stoma](#), or opening, in the [abdomen](#) that connects to your urinary tract to allow urine to drain freely from the body.
- Urine is collected and stored in a small bag, called a [urostomy pouch](#), which you can empty at convenience.
- **The pouch** is attached to the skin around the stoma and worn outside the body.

The two main types of urostomy include:

Ileal conduit.

- A surgeon removes a piece of the intestine to create a passage way for urine.
- The ureters are attached to the piece of intestine, then the intestine is attached to an opening in the abdomen, creating a stoma.
- The urine flows from the ureters, through the piece of intestine, and out the stoma.

Cutaneous ureterostomy.

A surgeon attaches one or both ureters directly to a stoma in the abdomen.

Continent Urinary Diversion

Continent urinary diversion collects and stores urine inside the body until the urine is drained using a catheter or through the urethra.

- The urine flows through the ureters and is stored in an internal pouch created from part of the [bowel](#) or in bladder.
- Continent urinary diversion allows you to control when urine leaves the body.

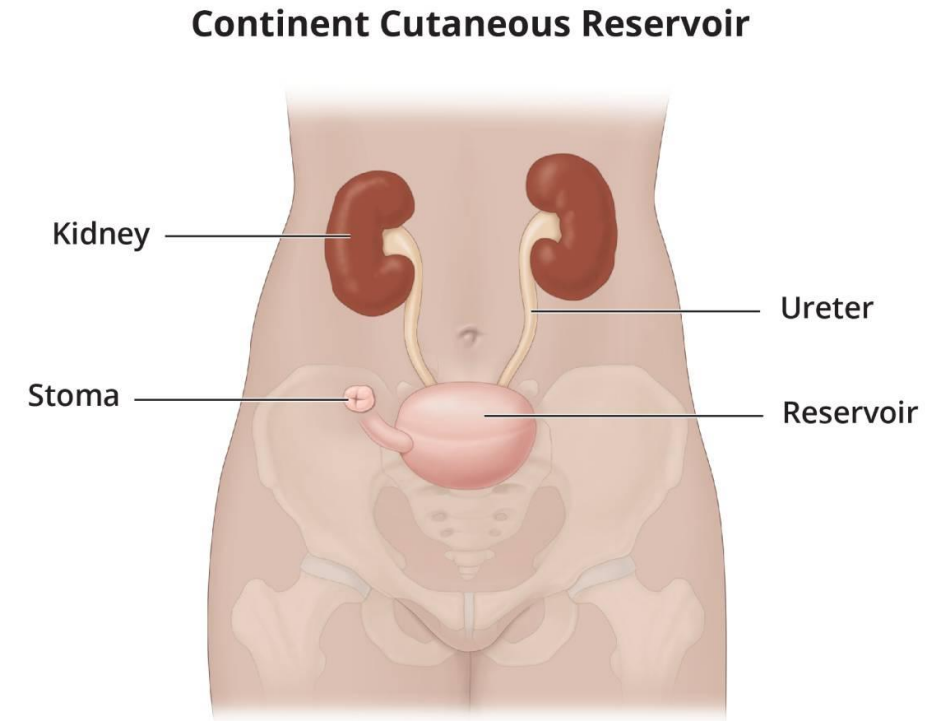
The main types of continent urinary diversion include:

1-Continent Cutaneous Reservoir.

2-Bladder Substitute

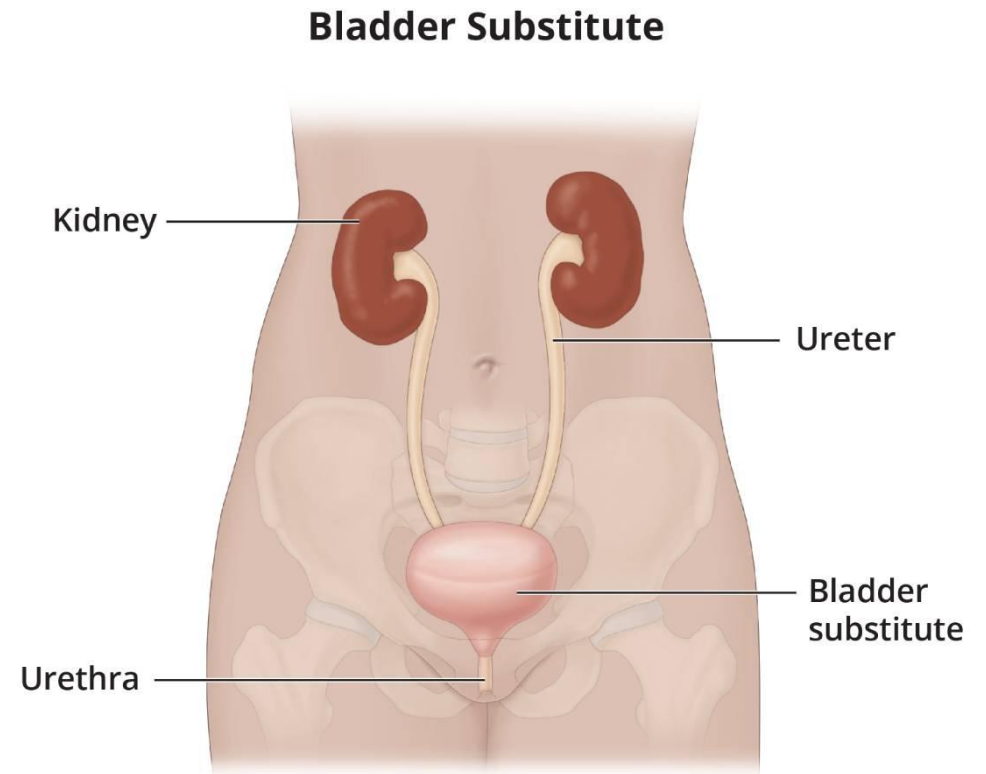
1-Continent Cutaneous Reservoir.

- A surgeon uses a piece of the bowel to create an internal pouch, or reservoir, to hold urine.
- The internal pouch is placed inside the abdomen.
- The ureters are attached to the internal pouch, and the internal pouch is attached to a stoma in the abdomen.
- Urine flows through the ureters and into the internal pouch, where it is stored until the patient drains the urine by inserting a catheter into the stoma.
- The stoma is the end of a channel that connects to the reservoir. The channel has a valve that prevents urine from exiting the body until a catheter is inserted.
- The channel can be created from a piece of intestine or by using the [appendix](#).



2-Bladder substitute, or neobladder.

- A surgeon uses a piece of the bowel to create an internal reservoir, called a bladder substitute or neobladder, to hold urine.
- The bladder substitute is placed in the [pelvis](#).
- The ureters are attached to the bladder substitute, and the bladder substitute is attached to the urethra.
- Urine flows through the ureters, into the bladder substitute, and the patient urinates through the urethra.



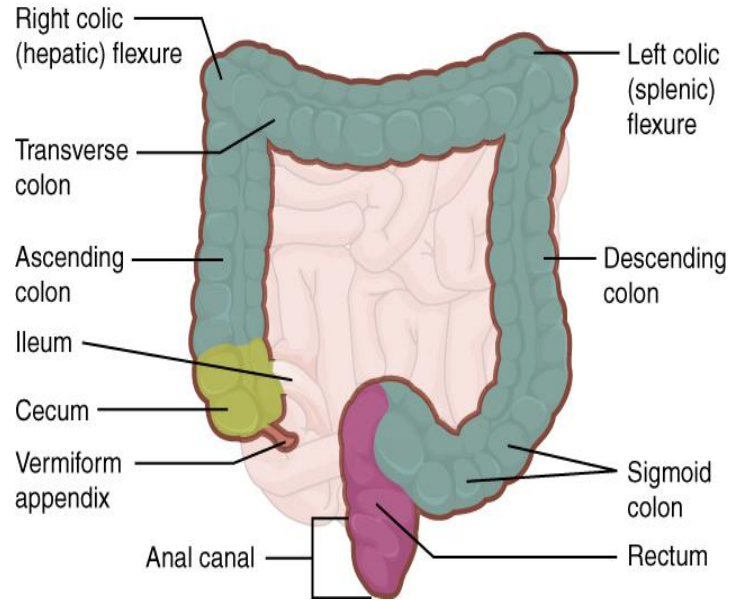
Caring for a stoma

Care for the stoma every day to keep the stoma and skin around the stoma clean and healthy.

- Gently wipe away mucus.
- **Wash** the stoma and surrounding skin with warm water.
- **Choose** mild soap only.
- **Rinse** the stoma and surrounding skin thoroughly.
- **Avoid** products that contain oils, fragrance, deodorants, alcohol, or harsh chemicals.
- **Inspect the stoma** and surrounding skin and contact health care professional if any skin changes or irritation are noticed.



Principles of Colon Surgery



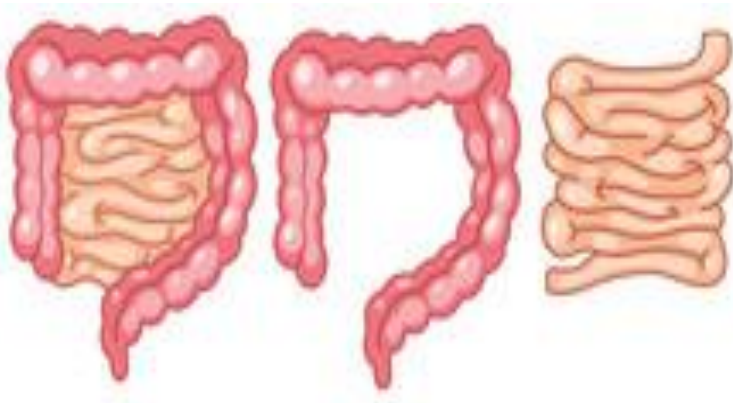
Colonic anastomosis is more liable to disruption, leakage and peritonitis more than small intestine due to:

1. The high **bacterial** count of both aerobic and anaerobic organisms.

2. Constant **gaseous** distension.

3. Incomplete serous coat.

4. The terminal **arteries** poorly connect with each other.



Pre-operative Preparation for Elective Surgery

1- Improving nutritional status.

2- Bowel preparation:

- The risk of anastomosis leakage and wound sepsis is reduced if the large bowel is empty at the time of resection and if the bacterial flora of the colon is reduced.

A- Mechanical preparation:

-Standard preparation:

- Low-residue diet for **4 days** before surgery.
- Enemas and laxatives for 2-3 days before surgery.

- Rapid preparation

This method is not used in cardiac or renal patients

- Alternative method can be performed **one day** before surgery.
- Whole gut irrigation using 2-4 liters / hour of a balanced **crystalloid** solution passed via NG tube until the patient passes clear fluid per rectum.

Bowel Preparation

B- Chemical preparation:

- Intestinal antiseptics (neomycin and metronidazole) for **2 days** before operation.
- Systemic parenteral antibiotics: given before surgery and are continued postoperatively for **1 day**.
- It is a combination of cephalosporin or an aminoglycoside with Metronidazole.

Operative Procedure

Resection:

- The extent of resection is governed by vascularity and by disease extent.
- In radical surgery for malignancy, it is essential to remove the draining lymphatics & LNs.

Anastomosis: 2 bowel ends should be:

- Adequately vascularized
 - Sutured without tension.
 - Water-tight & gas tight.
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Sutures

An intestinal anastomosis can be done by one of two methods:

1- Hand Suturing

- Is commonly done in **2 layers** of interrupted suture.
- The first layer (whole through): includes the whole wall thickness.
- The second layer (inverting Lambert): includes the serosa and muscle only.

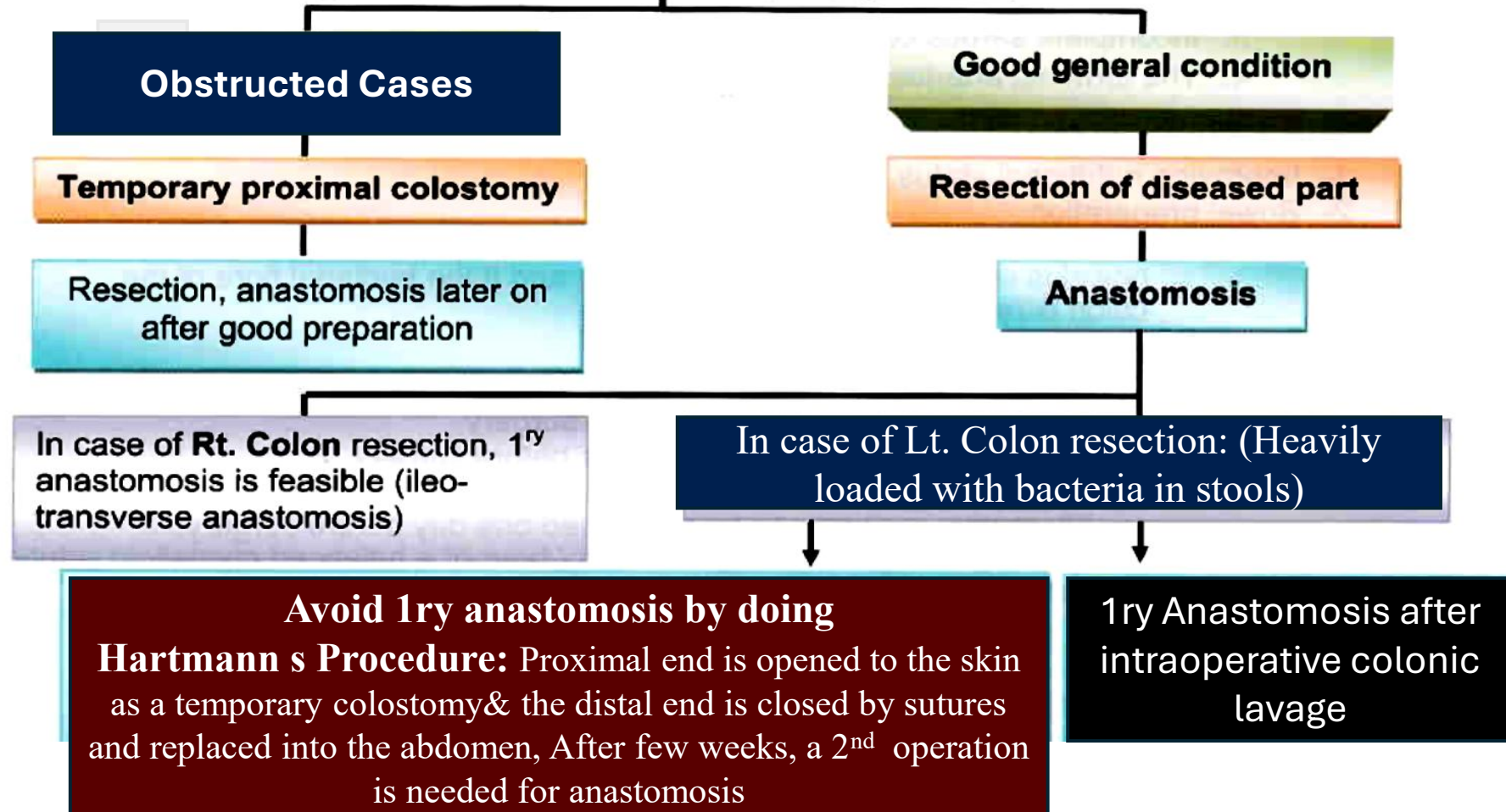
2. Mechanical Staplers: They are faster but more costly.

Emergency Surgery

Indications:

- Intestinal Obstruction
- Intestinal Perforation
- Toxic dilation
- Massive bleeding from the colon

➤ According to general condition of the patient:





Stoma Formation

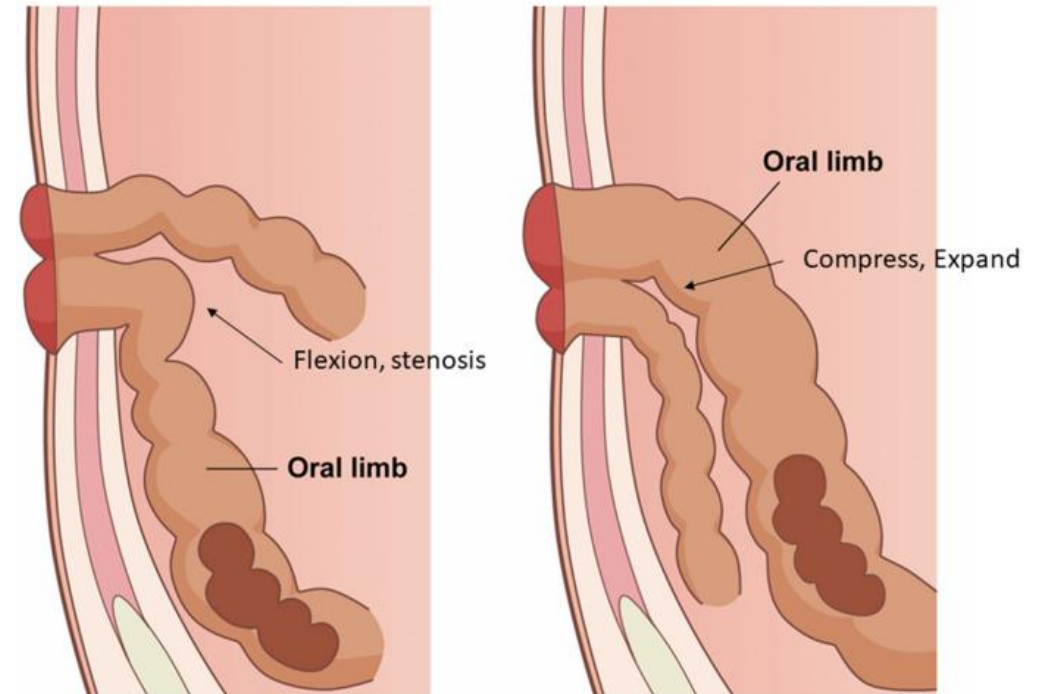
Pre-theatre preparation

- GA if done as part of other procedures or if laparoscopically formed; may be LA/regional if isolated loop stoma formation.
 - Blood group and save required.
 - Ideal sites for proposed stomas should be marked by stoma specialist preoperatively unless emergency surgery.
 - Single dose antibiotics.
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Steps of surgery

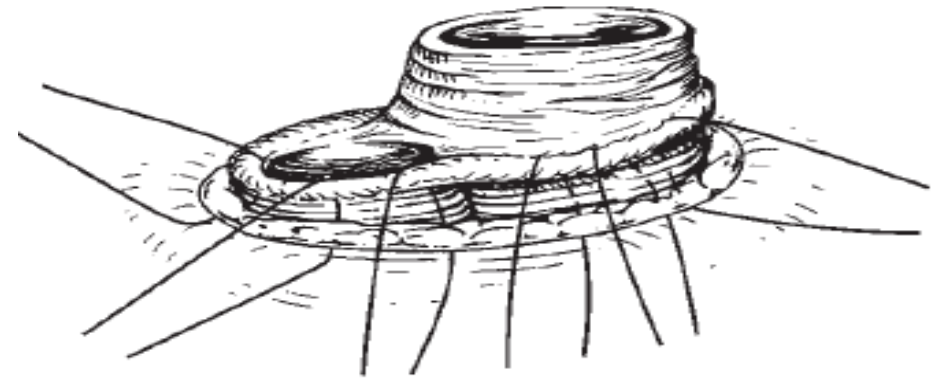
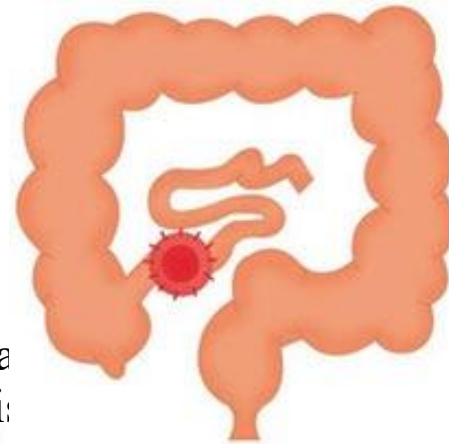
Positioning and theatre set-up

- **Table positioning:** Supine unless required by related operation.
- Ideally, any stoma should traverse the musculo-fascial layers of the abdominal wall (commonly the rectus abdominis); incisions in the fascia placed vertically with muscle fibres splayed apart (but not divided unless where necessary) to allow the stoma adequate space without 'pinching', but reduce the longer-term risk of herniation (a major problem especially in long-term stomas).



Loop ileostomy formation

- Ensuring proximal and distal limbs are identified and oriented (if done as an isolated procedure via a trephine incision, this is difficult; the only reliable indicator is the antemesenteric fat 'stripe' leading onto the surface of the caecum; much more reliably done laparoscopically for direct visualization of each limb).
- Opening of the antemesenteric wall of the loop (slightly oblique towards the distal limb).
- Placement of sutures (untied) to 'spout' the proximal limb.
- Placement and tying of sutures across the posterior wall of the loop and around the distal limb.
- Tying the proximal limb sutures and spouting the proximal limb.
- **Closure:** Port sites. As per laparoscopy.



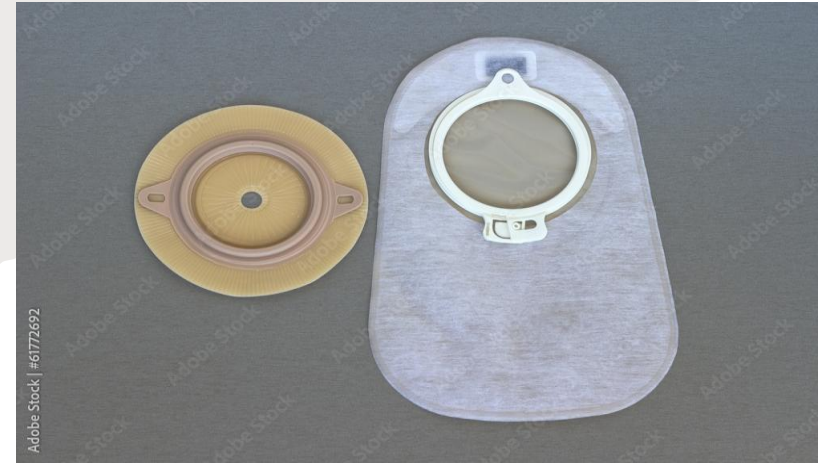
i Suture placement for spouting of proximal limb of loop ileostomy.

Post-operative Care and Instructions

-Oral diet: Normal as soon as tolerated unless indicated by further procedure performed.

Complications (specific to the stoma formation)

- Stomal oedema (common and usually requires no treatment).
- Stomal ischaemia, (serious and requires prompt treatment).
- Intestinal obstruction (due to tight musculo-fascial opening)
- Stoma retraction
- Stoma prolapse
- Stomal/parastomal hernias (rare acutely), up to 50% long term.



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Thank You